.3 immersion of any part of piping or ventilation ducts carried through a watertight boundary that is located within any compartment included in damage cases contributing to the attained index A , if not fitted with watertight means of closure at each boundary.

5.4 However, where compartments assume d flooded due to prog ressive flooding are taken into account in the damage stability calculations multiple values of sintermediate,i may be calculated assuming equalization in additional flooding phases.

5.5 Except as provided in paragraph 5.3.1, openings closed by means of watertight manhole covers and flush scuttles, small wate rtight hatch covers, remotely operated sliding watertight doors, side scuttles of the non-opening type as well as watertight access doors and hatch covers required to be kept closed at sea need not be considered.

6 Where horizontal watertight boundaries are fitted above the waterline under

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SOURCE: MSC.194(80).pdf
doors and hatch covers required to be kept closed at sea need not be considered.

6 Where horizontal watertight boundaries are fitted above the waterline under consideration the s-value calculated for the lower compartment or group of compartments shall be obtained by multiplying the value as determined in paragraph 1.1 by the reduction factor vm according to paragraph 6.1, which represents the probability that the spaces above the horizontal subdivision will not be flooded.

6.1 The factor vm shall be obtained from the formula:

$$vm = v(Hj, n, m, d) – v(Hj, n, m-1, d)$$

where:

Hj, n, m is the least height above the baselin e, in metres, within the longitudinal range of x1(j)...x2(j+n-1) of the mth horizontal boundary which is assumed to limit the vertical extent of flooding for the damaged compartments under consideration;

Hj, n, m-1 is the least height above the baseli ne, in metres, within the longitudinal range of x1(j)...x2(j+n-1) of the (m-1)th horizontal boundary which is assumed to limit the vertical extent of flooding for the damaged compartments under consideration;

j signifies the aft terminal of the damaged compartments under consideration;

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SOURCE: MSC.194(80).pdf
the vertical extent of flooding for the damaged compartments under consideration;

j signifies the aft terminal of the damaged compartments under consideration;

m represents each horizontal boundary counted upwards from the waterline under consideration;

d is the draught in question as defined in regulation 2; and

x1 and x 2 represent the terminals of the co mpartment or group of compartments considered in regulation 7-1.

6.1.1 The factors v(Hj, n, m, d) and v(Hj, n, m-1, d) shall be obtained from the formulae:

, if (Hm-d) is less than, or equal to, 7.8 m;

in all other cases,

()
8 . 78 . 0) , (d Hd H v =
()

+ = 7 . 4 8 . 72 . 0 8 . 0) , (d Hd H v

where:

v(Hj, n, m, d) is to be taken as 1, if Hm coincides with the uppermost watertight boundary of the ship within the range (x1(j)...x2(j+n-1)), and

v(Hj, n, 0, d) is to be taken as 0.

In no case is vm to be taken as less than zero or more than 1.

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SOURCE: MSC.194(80).pdf
boundary of the ship within the range (x1(j)...x2(j+n-1)), and

v(Hj, n, 0, d) is to be taken as 0.

In no case is vm to be taken as less than zero or more than 1.

6.2 In general, each contribution dA to the index A in the case of horizontal subdivisions is obtained from the formula:

() ( )[] mmi s vs v v s v p dA min 12 min 1 2 1 min 11 +++ =

where:

vm = the v-value calculated in accordance with paragraph 6.1;
smin = the least s-factor for all combinations of damages obtained when the assumed damage extends from the assumed damage height Hm downwards.

Regulation 7-3
Permeability

1 For the purpose of the subdivision and damage stability calculations of the regulations, the permeability of each general compartment or part of a compartment shall be as follows:

Spaces Permeability
Appropriated to stores 0.60
Occupied by accommodation 0.95
Occupied by machinery 0.85
Void spaces 0.95
Intended for liquids 0 or 0.951
1 Whichever results in the more severe requirement.

2 For the purpose of the subdivision and damage stability calculations of the

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SOURCE: MSC.194(80).pdf
Occupied by accommodation 0.95
Occupied by machinery 0.85
Void spaces 0.95
Intended for liquids 0 or 0.951
1 Whichever results in the more severe requirement.

2 For the purpose of the subdivision and damage stability calculations of the regulations, the permeability of each cargo comp artment or part of a compartment shall be as follows:

Spaces Permeability at draught ds Permeability at draught dp Permeability at draught dl
Dry cargo spaces 0.70 0.80 0.95
Container spaces 0.70 0.80 0.95
Ro-ro spaces 0.90 0.90 0.95
Cargo liquids 0.70 0.80 0.95

3 Other figures for permeability may be used if substantiated by calculations.
RESOLUTION MSC.194(80)
(adopted on 20 May 2005)
AMENDMENTS TO THE INTERNATIONAL CONVENTION FOR THE SAFETY OF LIFE AT SEA, 1974, AS AMENDED-27 -

Regulation 8
Special requirements concerning passenger ship stability

1 A passenger ship intended to carry 400 or more persons shall have watertight subdivision abaft the col lision bulkhead so that si = 1 for the three loading conditions on

which is based the calculation of the subdivision index and for a damage involving all the

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SOURCE: MSC.194(80).pdf
subdivision abaft the col lision bulkhead so that si = 1 for the three loading conditions on which is based the calculation of the subdivision index and for a damage involving all the compartments within 0.08L measured from the forward perpendicular.

2 A passenger ship intended to carry 36 or more persons is to be capable of withstanding damage along the side shell to an extent specified in paragraph 3. Compliance with this regulation is to be achieved by demonstrating that si, as defined in regulation 7-2, is not less than 0.9 for the three loading conditions on which is based the calculation of the subdivision index.

3 The damage extent to be assumed when demonstrating compliance with paragraph 2, is to be dependent on both N as defined in regulation 6, and Ls as defined in regulation 2, such that:

.1 the vertical extent of damage is to extend from the ship’s moulded baseline to a position up to 12.5 m above the position of the d eepest subdivision draught as defined in regulation 2, unless a lesser vertical extent of damage were to give a lower value of si, in which case this reduced extent is to be used;

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